

ETHAN (TAEHOON) LEE

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Software Engineering (4.0/4.0 GPA)

Expected Dec 2027

- **Courses:** Distributed Systems, Operating Systems, Cloud Computing, System Design

Emory University

Atlanta, GA

BS in Computer Science (4.0/4.0 GPA, Summa Cum Laude, 5x Dean's List)

May 2026

- **Courses:** Data Structures & Algorithms, Computer Architecture, Databases, Object-oriented Programming

WORK EXPERIENCE

Software Engineer

Atlanta, GA

Huddle Surety

Sep 2025 – Present

- Built an AI platform replacing manual surety bond processing at 10x speed, running 100+ bonds daily across 5 agencies
- Designed and developed the core Go REST API spanning 100+ endpoints with Redis caching and Temporal orchestration
- Engineered Python LLM retrieval (RAG) pipelines with cross-document context, generating bond forms at 95% accuracy
- Shipped a role-based web frontend delivering distinct surety agent and contractor UX in React, Next.js, and TypeScript
- Architected end-to-end distributed tracing with OpenTelemetry, enabling per-client usage metrics and cost analysis

Software Engineering Intern (Cloud Infrastructure)

Atlanta, GA

UKG

May 2025 – Aug 2025

- Automated infrastructure operations in a hybrid cloud environment spanning 4 on-prem datacenters and 3 GCP regions
- Built a TLS certificate lifecycle monitor in Go, packaged as a Helm chart and scheduled as a Kubernetes CronJob, replacing manual healthchecks of 200+ Linux VMs and cutting annual certificate renewal time by 50%
- Partnered with Grafana Labs engineers to build database synthetic monitoring using Grafana K6 and Blackbox Exporters
- Developed a Go REST API using concurrent scatter-gather to aggregate network metrics across 5 Kubernetes clusters

Software Engineering Intern (Full Stack)

Atlanta, GA

eStreamly

Jun 2024 – Aug 2024

- Cut redundant API calls by 40% with client-side caching and reactive state store in livestream shopping platform video UI
- Created a multi-stage Docker build and CI/CD release pipeline on GCP Cloud Build, improving deployment speed by 35%

PROJECTS

Kubernetes Homelab & Contribution | Kubernetes, Linux, Operating System, CI/CD

headlamp.thdxg.dev

- Operated a Kubernetes cluster on 2 Raspberry Pi Linux nodes, serving 20+ projects including personal website
- Automated GitOps CI/CD and provisioned public ingress using Cilium container networking interface and Gateway API
- Contributed upstream to Kubernetes as a SIG Docs member, authoring Korean documentation on scheduling and eviction

Macterm: Terminal Emulator with Persistent Multiplexing | Swift, SwiftUI, C

github.com/thdxg/macterm

- Built and maintain a native macOS terminal in Swift and C, with 8K+ downloads and 20 global open-source contributors
- Implemented session persistence, declarative layout configuration, fuzzy command palette, and configurable hotkeys

DooleyOnline: Emory Student Marketplace | Go, Next.js, React, TypeScript, PostgreSQL, AWS

github.com/dooleyonline

- Launched a campus marketplace of 150+ users with a Go, PostgreSQL backend and Next.js, TypeScript, React frontend
- Implemented real-time buyer-seller chat over WebSockets with persisted message history and S3-backed image uploads

Spotify Warped: HCI Research & Publication | Python, JavaScript, React

dl.acm.org/doi/10.1145/3772318.3790875

- Published at ACM CHI 2026 on Spotify's algorithmic feedback loops, building Python pipelines over 100 GB of user data

SKILLS

- **Languages:** Go, Python, TypeScript, Java, C, JavaScript, Rust, SQL, Swift, x86 Assembly
- **Technologies:** Kubernetes, Docker, Linux, AWS, GCP, PostgreSQL, Redis, Git, Helm, OpenTelemetry, Prometheus, Grafana